

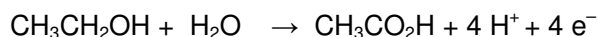
Section A

Answer **all** questions in this section in the spaces provided

For
Examiner's
use

- 1 Drink driving is a common problem observed during festive seasons. To detect the drivers who have violated the law, the traffic police would carry with them a portable breathalyser to test for alcohol content from the suspect's breath. One of the first chemical breathalysers was based on the oxidation process of ethanol using acidified potassium dichromate(VI).

- (a) Given the half equation for the oxidation of ethanol, using relevant [1]
information from the *Data Booklet*, write a balanced equation for oxidation of ethanol using acidified potassium dichromate(VI).



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- (b) What colour change would you expect for a positive test? [1]

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The ethanol content in a can of beer with a volume of 330 cm³ can be determined in the laboratory by titrating a sample of beer with potassium dichromate(VI) acidified by dilute sulfuric acid.

A 25.0 cm³ sample of beer was pipetted from a brand new can to a 250 cm³ volumetric flask. Deionised water was added to the flask until the 250 cm³ mark. 25.0 cm³ of the diluted beer was then pipetted out and it required 17.15 cm³ of 0.050 mol dm⁻³ of acidified potassium dichromate(VI) for complete titration.

- (c) Calculate the number of moles of ethanol titrated. [2]

(d) Calculate the number of moles of ethanol present in one can of beer.

[2]

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(e) Hence calculate the concentration of ethanol in the can of beer.

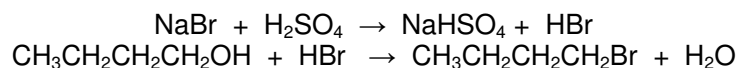
[1]

[Total: 7]

2 1-bromobutane may be made by reacting together butan-1-ol, sodium bromide and concentrated sulfuric acid. In one such synthesis,

- 35 g of sodium bromide was mixed with 25 cm³ of butan-1-ol and 30 cm³ of water in a flask.
- 25 cm³ of concentrated sulfuric acid was then added dropwise into the flask containing the reagents.
- As the acid was added the contents were well shaken and cooled occasionally in an ice bath.
- Once all the acid has been added, the mixture was gently boiled for some time shaking the flask gently every now and then.
- The apparatus was then rearranged to distil off the 1-bromobutane

The reaction equations are as follows:



Substance	Boiling point / °C	Density / g cm ⁻³	M _r
1-bromobutane	102	1.35	137
Butan-1-ol	118	0.81	74
NaBr	1390	3.20	103
Concentrated H ₂ SO ₄	330	1.84	98

- (a) Despite the larger M_r , 1-bromobutane has a lower boiling point than butan-1-ol. Why? [2]

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- (b) Based on the amounts used in the synthesis above, one of the reagents, butan-1-ol or sodium bromide will be present in an excess. Use the data above to determine which reagent is in excess. [2]

- (c) After distillation and purification, 20 cm³ of 1-bromobutane was collected. Calculate the percentage yield in the above preparation. [2]

- (d) In the light of your answer to (c) suggest why is the percentage yield not 100%? [1]

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- (e) During the dropwise addition of concentrated sulfuric acid to the reaction mixture, occasional cooling was necessary. What does this suggest about the nature of reaction taking place and why is cooling necessary? [2]

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- (f) What type of reaction took place between butan-1-ol and the HBr generated in the first step? [1]

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- (g) In the light of your answer to (f), why was it necessary for the mixture to be boiled for some time? [1]

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- (h) By means of a simple chemical test, how will you distinguish between 1-bromobutane and butan-1-ol. [1]

[Total: 12]

- 3 Silicon is a tetravalent metalloid, which is less reactive than carbon, but more reactive than germanium. The element has a large impact on the modern world economy, as it is used in chemical and semiconductor industries. Silicon is also an essential element in biology and is of much importance to the metabolism of plants.

- (a) (i) The atomic radii of silicon and aluminium are given as follows: [3]

Element	Silicon	Aluminium
Atomic radius/ nm	0.117	0.143

Account for the difference between the atomic radius of silicon and aluminium.

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- (ii) The melting points of silicon and phosphorus are given as follows:

Element	Silicon	Phosphorus
Melting point/ °C	1414	44.2

Account for the difference in the melting points of the 2 solids.

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- (b) Silicon tetrachloride is used as an intermediate in the manufacture of high purity silicon. Silicon derived from silicon tetrachloride is used in the semiconductor industry and also the production of photovoltaic cells. [3]

- (i) Illustrate the formation of silicon tetrachloride from its elements with an equation, including state symbols.

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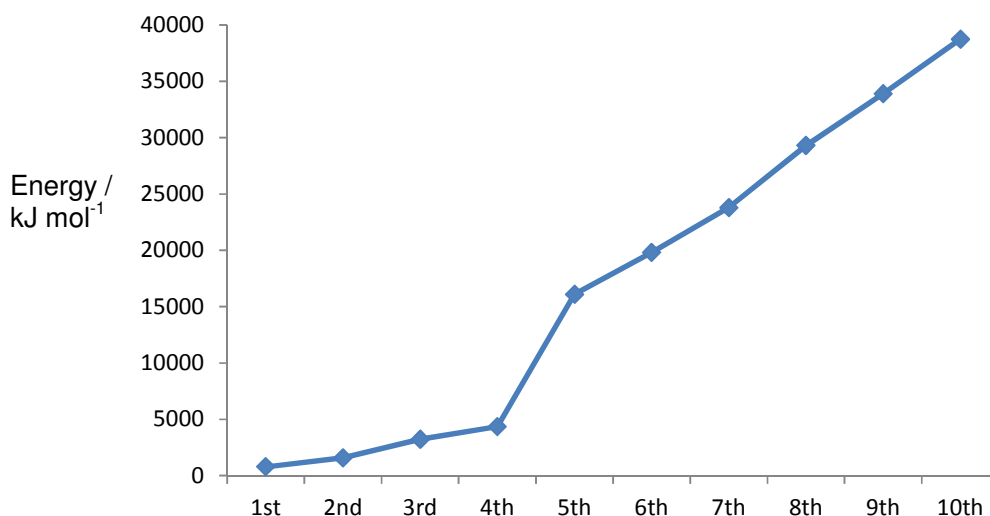
- (ii) Describe, with the aid of a suitable equation, how and why silicon tetrachloride is able to react with water.

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- (c) The following graph shows a plot of the successive ionisation energies (IE) of silicon: [2]



- (i) Account for the general increase in successive ionisation energies.

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- (ii) Account for the large increase between 4th and 5th ionisation energies.

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[Total: 8]

- 4 (a) 1-iodopropane can be hydrolysed by NaOH(aq) to form propan-1-ol. Results [5]
of an investigation into the kinetics of this reactions are given below.

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Experiment number	Concentration / mol dm ⁻³		Relative initial rate
	1-iodopropane	OH ⁻	
1	0.10	0.20	1.00
2	0.20	0.40	4.00
3	0.30	0.20	3.00
4	0.40	0.40	8.00

- (i) Determine the order of reaction with respect to 1-iodopropane and OH⁻.

- (ii) Hence give the rate equation for the hydrolysis of 1-iodopropane.

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- (iii) Draw a fully labeled energy profile diagram for this hydrolysis.

- (b) 15 drops of 1-chloropropane, 1-bromopropane and 1-iodopropane was put in 3 separate test tubes. Dilute silver nitrate solution was added to each test tube and all 3 test tubes were heated in a water bath. The following observations were obtained. [2]

Test tube	Colour of ppt	Time taken
1	Cream	After 8 minutes
2	White	After 20 minutes
3	Yellow	After 1 minute

- (i) Identify the halogen derivative in each test tube.

Test tube 1:

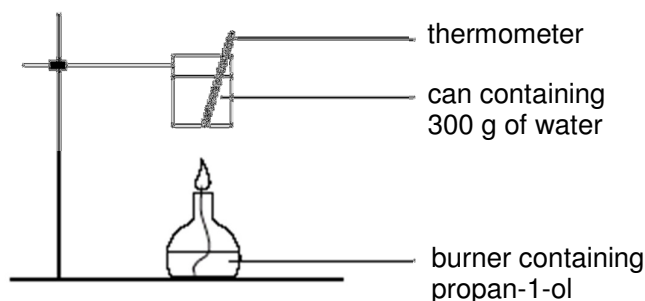
Test tube 2:

Test tube 3:

- (ii) Explain why the time taken in all 3 test tubes is different.

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- (c) A student attempted to determine the enthalpy change of combustion of propan-1-ol in the laboratory. She set up the experiment according to the diagram given below: [6]



The following data were recorded:

Volume of water in beaker = 300 cm³

Initial temperature of water = 27 °C

Final temperature of water = 37 °C

Initial mass of spirit lamp with propan-1-ol = 158.65 g

Final mass of spirit lamp with propan-1-ol = 158.25 g

- (i) Using the information given above, calculate the enthalpy change of combustion of propan-1-ol.

(Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

(M_r of propan-1-ol = 60.0)

- (ii) The student compared the enthalpy change of combustion that she obtained to the value found in the textbook. The value given in the textbook is $-2020 \text{ kJ mol}^{-1}$. Suggest a reason for the difference.

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- (iii) Use the bond energies given in the *Data Booklet* to calculate another value for the standard enthalpy change of combustion of propan-1-ol.

- (iv) Suggest a reason for the discrepancy between this value and that quoted in (c)(ii).

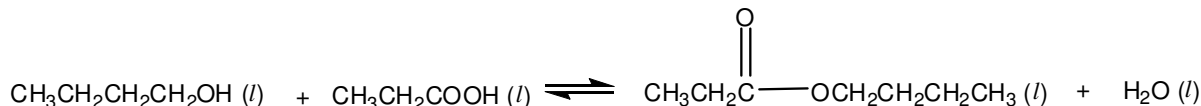
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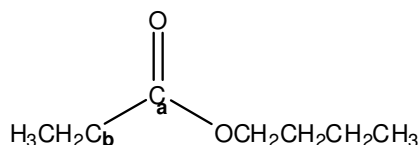
Section B

Answer **two** questions from this section on separate answer paper

- 5 (a) It was discovered that butylpropanoate is a commonly used ester in food flavouring. [9]
This ester can be made in the laboratory by heating butan-1-ol and propanoic acid with trace amount of concentrated sulfuric acid according to the equation given.



- (i) Concentrated sulfuric acid plays 2 roles in the above reaction. What are the 2 roles?
- (ii) Write the K_c expression.
- (iii) The synthesis of butylpropanoate was done using the method suggested above. At equilibrium, the concentration of butan-1-ol and propanoic acid was found to be $0.480 \text{ mol dm}^{-3}$ and butylpropanoate and water was found to be 1.32 mol dm^{-3} . Calculate the value of K_c .
- (iv) State and explain what would happen to the position of equilibrium and K_c value if more butan-1-ol is added to the reaction mixture.
- (v) The structure of butylpropanoate is given below. State the hybrid state of C_a and C_b , and shape and angle around C_a and C_b .



- (b) Butan-2-ol can be formed from butan-1-ol via this pathway in the laboratory: [4]



- (i) Give the reagents and conditions needed for Step I and II.
 - (ii) Draw the structure of **A**.
 - (iii) State and explain whether **A** can exhibit geometrical isomerism.
- (c) **B** and **C** are isomers with molecular formula $\text{C}_4\text{H}_{10}\text{O}$. **B** and **C** react with sodium to give a gas which extinguishes a lighted splint with a “pop” sound. When heated with acidified manganate(VII), **B** and **C** decolourise manganate(VII) to give **D** and **E** respectively. **D** does not react with NaHCO_3 but gives an orange precipitate upon adding 2,4-dinitrophenylhydrazine and yellow precipitate with aqueous alkaline iodine. **E** reacts with NaHCO_3 giving a colourless and odourless gas. **C** reacts with hot concentrated H_2SO_4 to give an alkene. The alkene decolorizes hot acidified manganate(VII) to give a neutral organic compound with the formula $\text{C}_3\text{H}_6\text{O}$. [7]
Deduce the structures of **B**, **C**, **D** and **E** with reasoning.

[Total: 20]

- 6 Aluminium is the third most abundant element and most abundant metal in the Earth's crust. Aluminium is highly sought after and has many commercial uses in the aerospace and construction industries, due to its low density and its ability to resist corrosion.

- (a) Aluminium oxide is produced or extracted from minerals on a large scale. It is a common catalyst for many industrial processes and also widely used as an abrasive due to its hardness. [2]

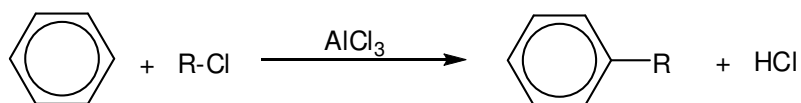
Explain, with the aid of equations, the amphoteric properties of aluminium oxide.

- (b) Some of the common compounds of aluminium are given as follows: [4]



- (i) Arrange the compounds of aluminium in increasing order of melting point.
- (ii) Explain your answer in (b)(i), in relation to the structure and bonding of the aluminium compounds.
- (c) Aluminium chloride is used in petroleum refining and in the production of synthetic rubber and polymers. Its anhydrous form is a powerful Lewis acid, which is an electron-pair acceptor, and reacts vigorously with bases, hence suitable precautions must be taken when handling this compound. [5]
- (i) Draw a dot-and-cross diagram showing the arrangement of outer shell electrons in AlCl_3 . Suggest the shape of the molecule and the bond angle about the central atom.
- (ii) Draw the structure of AlCl_3 upon dimerization. Hence, suggest if all of the chlorine atoms lie on the same plane.
- (d) Friedel-Crafts alkylation reactions are developed in 1877 to attach alkyl-substituents to an aromatic ring. These reactions are usually catalysed by anhydrous aluminium chloride. [9]

The following equation illustrates a Friedel-Craft alkylation reaction:



- (i) Benzene reacts with compound **F**, in the presence of anhydrous AlCl_3 to form compound **G**, C_8H_{10} .

Deduce the structure and physical state of compound **F**.

- (ii) (1) Oxidation of compound **G** in the presence of hot $\text{MnO}_2(\text{s})$ produces compound **H**, $\text{C}_7\text{H}_6\text{O}$, which forms an orange precipitate with 2,4-dinitrophenylhydrazine.

Compound **J** is formed when compound **G** is reacted with chlorine gas in the presence of UV light. When **J** is heated under reflux with $\text{NaOH}(\text{aq})$, compound **K** is formed. When **K** is heated at high temperatures and pressure with $\text{Al}_2\text{O}_3(\text{s})$, compound **L** is formed.

When **K** is heated with acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, compound **M** is formed, which

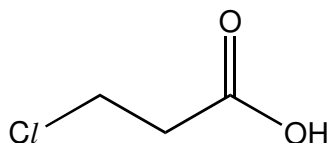
also forms an orange precipitate with 2,4-dinitrophenylhydrazine.

Deduce the structures of compounds **H**, **J**, **K**, **L** and **M**.

- (2) Construct a balanced equation to illustrate the reaction between compound **M** and 2,4-dinitrophenylhydrazine.
- (3) By means of a simple chemical test, suggest how you could distinguish between compounds **H** and **M**.

[Total: 20]

- 7 (a) 3-chloropropanoic acid is a drug used in scientific research due to its structural [4]
resemblance to 4-hydroxybutanoic acid, a naturally occurring substance found in the
central nervous system.



3-chloropropanoic acid

- (i) The pK_a of propanoic acid is 4.87. Predict, with reasons, how the pK_a of 3-chloropropanoic acid would differ from propanoic acid.
 - (ii) A positional isomer of 3-chloropropanoic acid is 2-chloropropanoic acid. Compare the acid strength of 2-chloropropanoic acid with 3-chloropropanoic acid. Explain your answer.
- (b) 0.020 mol of 3-chloropropanoic acid was dissolved in 50.0 cm³ of water to produce **FA1** [9]
which was then titrated with **FA2**, a 0.50 mol dm⁻³ solution of NaOH. After
20.0 cm³ of NaOH was added, the pH of the solution was found to be 3.98.
- (i) Determine the pK_a of 3-chloropropanoic acid.
 - (ii) The resultant mixture from (b) acts as a buffer mixture. By means of balanced equations illustrate how this mixture acts as a buffer.
 - (iii) Calculate the pH of the solution **FA1**.
 - (iv) What will be the pH of the solution mixture at the equivalent point if **FA1** was neutralized completely with **FA2**.
- (c) Outline how you would synthesise 2-chloropropanoic acid starting from ethanal, stating [5]
reagents and conditions for each step of the conversion.
- (d) By means of a simple chemical test, suggest how you could distinguish between [2]
3-chloropropanoic acid and propanoic acid.

[Total: 20]